Lecture 18, Math 441

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Linear Inhomogeneous IVP

$y''' - y' = 6e^{2x} \qquad y(0) = 2; y'(0) = 0; y''(0) = 2$

1. Find y_c, a complementary solution to the homogeneous equation.
2. Find y_p, a particular solution to the inhomogeneous equation.
3. Add $y = y_c + y_p$ to get a general solution of the inhomogeneous equation.
4. Apply initial conditions to $y = y_c + y_p$, and find the solution to the IVP.

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Method of Undetermined Coefficients

Find a particular solution y_p for inhomogeneous linear

constant coefficient equation.

Ex: $y'' - y = e^{2x}$

$y_p = A\,e^{2x}$

$y_p' = 2\,A\,e^{2x}, \quad y_p'' = 4\,A\,e^{2x}$

Substituting: $4\,A\,e^{2x} - A\,e^{2x} = e^{2x} \;\rightarrow\; 3A = 1 \;\rightarrow\; A = \frac{1}{3}$

$$\boxed{y_p = \frac{1}{3}\,e^{2x}}$$

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Ex: $y' + y = x^2$

$y_p = A x^2 + Bx + C$

$y_p' = 2Ax + B$

$$\underbrace{2\,A\,x + B}_{y_p} + \underbrace{A\,x^2 + B\,x + C}_{y_p'} = x^2$$

Matching the coefficients of each power of x we get the three equations: $A = 1$; $2\,A + B = 0$; $B + C = 0$.

So, $A = 1$; $B = -2$; $C = 2$;

$$\boxed{y_p(x) = x^2 - 2\,x + 2}$$

Ex: $y'' + y = \cos(2x)$

Try $y_p = A\cos(2x)$

$y_p'' = -4A\cos(2x)$

Substituting:

$-4A\cos(2x) + A\cos(2x) = \cos(2x) \;\rightarrow\; A = -\frac{1}{3}$

and a particular solution is $\boxed{y_p = -\frac{1}{3}\cos(2x)}$

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Ex: $y'' + y' + y = \cos(2x)$

$y_p(x) = A\cos(2x) + B\sin(2x)$

$y_p' = -2A\sin(2x) + 2B\cos(2x)$

$y_p'' = 4A\cos(2x) + 4B\sin(2x)$

The equation is now:

$-4A\cos(2x) - 4B\sin(2x) - 2A\sin(2x) + 2B\cos(2x) + A\cos(2x) +$

$B\sin(2x) = \cos(2x)$

Matching all the coefficients for $\cos(2x)$ and $\sin(2x)$ separately:

$4A + 2B + A = 1$ from the $\cos(2x)$ terms.

$4B - 2A + B = 0$ from the $\sin(2x)$ terms.

Solving for A and B: $A = -\frac{3}{13}$; $B = \frac{2}{13}$

$y_p(x) = -\frac{3}{13}\cos(2x) + \frac{2}{13}\sin(2x)$

Ex: $\cdots = x\,\mathrm{e}^{3x}$

We should use $y_p(x) = A\,x\,\mathrm{e}^{3x} + B\,\mathrm{e}^{3x}$ and find A and B.

Ex: $\cdots = x^2\,\cos(x)$

We should use $y_p = A\,x^2\,\cos(x) + B\,x^2\,\sin(x) + C\,x\,\cos(x) +$

$D\,x\,\sin(x) + E\,\cos(x) + F\,\sin(x)$ and find $A, B, C, D, E,$ and F.

Method of Undetermined Coefficients (draft)

If the left hand side is constant coefficient and the right hand side is an exp., sin. or cos., or a polynomial, or a product of them, then try a y_p that has the same form, add all possible derivatives (each with a coefficient), and determine the coefficients.

$$0 = \cos(x)$$

$$y' + y = x^2$$

$$y_p = Ax^2$$

$$2Ax + Ax^2 = x^2$$

$$A = 1$$

$$2 = 0 \qquad (!)$$

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Ex: $y' - y = e^x$

$y_p = A e^x$; $y_p' = A e^x$

$A e^x - A e^x = e^x \rightarrow 0 = e^x$. This doesn't work.

That's because e^x is actually a term in the complementary solution

$y_c = c_1 e^x$:

$y' - y = 0 \qquad y = e^{rx} \qquad r - 1 = 0 \rightarrow r = 1 \rightarrow y_1 = e^x \rightarrow y_c = c_1 e^x$

$y_p = Axe^x$; $y_p' = Ae^x + Axe^x$

$Ae^x + Axe^x - Axe^x = e^x \rightarrow Ae^x = e^x \rightarrow A = 1 \rightarrow \boxed{y_p = xe^x}$

A general solution is $y = y_c + y_p = \boxed{c_1 e^x + x e^x}$

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Duplication between y_c and y_p

$y' - y = \mathrm{e}^x$

Rewrite as $(\mathrm{D} - 1)y = \mathrm{e}^x$.

Let's find a solution of the form $y(x) = v(x)\,\mathrm{e}^x$. Putting this in the equation we get: $(\mathrm{D} - 1)(v\,\mathrm{e}^x) = v'\,\mathrm{e}^x + v\,\mathrm{e}^x - v\,\mathrm{e}^x = v'\mathrm{e}^x$

$(\mathrm{D} - 1)v\mathrm{e}^x = \mathrm{e}^x$

$v'\mathrm{e}^x = \mathrm{e}^x$

$v' = 1 \;\rightarrow\; v = x + c$

Putting this back in $y = v\,\mathrm{e}^x$: $y = \underbrace{x\,\mathrm{e}^x}_{y_p} + \underbrace{c\,\mathrm{e}^x}_{y_c}$.

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Ex: $y'' - 3y' + 2y = 5e^x$

1. Find y_c. $y'' - 3y' + 2y = 0$

   $y = e^{rx}$

   $r^2 - 3r + 2 = 0 \rightarrow r = 1; r = 2$. So, $y_c = c_1 e^x + c_2 e^{2x}$

2. Find y_p. $y_p = Ae^x x$.

   $y_p' = \cdots$; $y_p'' = \cdots$

   Find A.

3. $y = y_c + y_p$

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